Public datasets



MIDV-2020: 10 designs,
scans and phone photos



SIDTD: forged versions,
clean source, altered field



FantasyID: 13 designs,
10 scripts, printed cards

Data preparation

- **Unified records: design, identity, label, type, fields, mask**
- **Rectify every page to its design canvas (512 x 800)**
- **Design median reference + zone mask per design**
- **Extra forgeries (SIDTD generator) with exact masks**
- **Pairs (page, reference, zone); identity-grouped splits**

Proposed architecture

- **Base: ChangerEx (Open-CD) Siamese change detection**
- **Standard: difference fusion |f(page) - f(ref)| + f(page) + zone**
- **Improved: attention fusion self-attn, cross-attn (Q = page, K,V = ref)**
- **Heads: alteration mask M and referral score s**
- **Grid: 4 backbones x 4 fusions x 2 references x 4 transfer regimes**

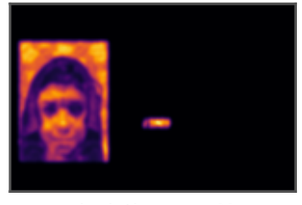
Training and evaluation

- **Loss: BCE(s) + 3 (BCE + Dice)(M) AdamW, early stopping, 3 seeds**
- **Augment: misalignment, jitter, independent JPEG on both inputs**
- **Protocol A: unseen design, SIDTD split**
  Protocol B: unseen dataset and script, FantasyID test (Arabic design alone)
- **Anchors: SIDTD baselines, TruFor zero-shot, same-capacity baselines**
- **Metrics: AUC, EER, APCER@BPCER, pixel F1; referral rate at miss 1, 5 %**

Output



presented page +
design reference



deviation map M

refer / clear at miss budget

released: data, splits, code